Boss Challenge — Paper 55 (Class 7)

Heat | Respiration in Organisms | Algebraic Expressions | Data Handling

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

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1. To find out exactly how hot a cup of milk is, you would use an instrument that measures temperature. That instrument is called a:

- **(A)** ruler
- **(B)** weighing scale
- **(C)** barometer
- **(D)** thermometer

2. Living things take in a gas from the air to break down food and release energy, and give out a different gas. The gas taken IN is:

- **(A)** hydrogen
- **(B)** nitrogen
- **(C)** carbon dioxide
- **(D)** oxygen

3. In algebra a letter such as x that can stand for many different numbers at different times is known as a:

- **(A)** equation
- **(B)** variable
- **(C)** coefficient
- **(D)** constant

4. The marks of five students are 6, 8, 10, 7 and 9. To find their AVERAGE (mean) mark, you would:

- **(A)** add all the marks and divide by 2
- **(B)** pick the largest mark
- **(C)** subtract the smallest from the largest
- **(D)** add all the marks and divide by 5

5. Hold a steel spoon in a cup of hot tea for a minute and the far end of the handle slowly turns warm. Heat has travelled along the solid metal by:

- **(A)** conduction
- **(B)** convection
- **(C)** radiation
- **(D)** evaporation

6. When you run fast and your muscles do not get enough oxygen, they break down sugar without it and a substance builds up that makes the muscles ache. That substance is:

- **(A)** lactic acid
- **(B)** glucose
- **(C)** oxygen
- **(D)** water

7. A number that has a fixed, unchanging value - such as the 9 in the expression x + 9 - is called a:

(A) variable
(B) term
(C) constant
(D) coefficient

8. In a set of numbers, the value that appears MORE OFTEN than any other is called the:

(A) mean
(B) mode
(C) median
(D) range

9. On a sunny day a person wearing a BLACK shirt feels hotter than a person in a WHITE shirt, because dark surfaces:

(A) absorb more heat
(B) make their own heat
(C) reflect more heat
(D) block all heat

10. A fish kept in a pond does not gulp air the way we do. It takes in the oxygen that is dissolved in the water using its:

(A) spiracles
(B) skin
(C) lungs
(D) gills

11. In the term 7y, the number 7 that is multiplied by the variable y is called the:

(A) exponent
(B) variable
(C) coefficient
(D) constant

12. To find the MEDIAN of an odd number of values, you first put them in order from smallest to largest and then pick the:

(A) most common value
(B) middle value
(C) largest value
(D) average of all values

13. The temperature of a healthy human body, marked on a clinical thermometer, is normally about:

(A) 37°C
(B) 50°C
(C) 0°C
(D) 100°C

14. When we breathe IN, a dome-shaped sheet of muscle below the lungs flattens and moves down, making more room for air. This sheet of muscle is the:

(A) gills
(B) ribs
(C) diaphragm
(D) windpipe

15. A class breathes about 18 times each minute. Written as an algebraic expression, the number of breaths taken in m minutes is:

(A) 18
(B) 18m
(C) m / 18
(D) 18 + m

16. The daytime temperatures for four days were 30°C, 32°C, 28°C and 30°C. The MEAN (average) temperature for these four days is:

(A) 28°C
(B) 120°C
(C) 30°C
(D) 32°C

17. Heat from the Sun reaches the Earth across empty space where there is no air. The Sun's heat must therefore travel by:

(A) boiling
(B) conduction
(C) radiation
(D) convection

18. Green plants are alive and so they respire all the time, taking in oxygen and giving out carbon dioxide. In a leaf, this gas exchange happens mainly through tiny pores called:

(A) petals
(B) veins
(C) stomata
(D) roots

19. Terms that have exactly the same variable part - for example 4xy and 9xy - are called:

(A) unlike terms
(B) equations
(C) constants
(D) like terms

20. A set of numbers is 4, 9 and 2. The RANGE of this data - the gap between the largest and smallest value - is:

(A) 11
(B) 5
(C) 2
(D) 7

21. A laboratory thermometer can measure a much wider range of temperatures than a clinical one. A typical laboratory thermometer reads from about:

- **(A)** -10°C to 110°C
- **(B)** 0°C to 50°C
- **(C)** 35°C to 42°C
- **(D)** 100°C to 500°C

22. An earthworm has no lungs and no gills. It takes in oxygen straight through its body surface, which works only as long as that surface stays:

- **(A)** coloured
- **(B)** moist
- **(C)** hard
- **(D)** dry

23. The expression for a number y that is 'increased by 5' - that is, 5 more than y - is written as:

- **(A)** $y - 5$
- **(B)** $5y$
- **(C)** $y / 5$
- **(D)** $y + 5$

24. The numbers 2, 4 and 6 have a mean of 4. If you now ADD the value 8 to this set, the new mean will be:

- **(A)** 5
- **(B)** 8
- **(C)** 4
- **(D)** 6

25. We feel warmer wearing a woollen sweater in winter mainly because wool traps a lot of tiny pockets of air, and trapped air is a:

- **(A)** poor conductor of heat
- **(B)** good conductor of heat
- **(C)** liquid
- **(D)** source of heat

26. When we breathe out and bubble the air through clear lime water, the lime water turns milky. This simple test shows that exhaled air is rich in:

- **(A)** water vapour
- **(B)** oxygen
- **(C)** nitrogen
- **(D)** carbon dioxide

27. To find the VALUE of the expression $2x + 3$ when $x = 4$, you replace x with 4. The value is:

- **(A)** 14
- **(B)** 24
- **(C)** 9
- **(D)** 11

28. A pictograph uses a small picture to stand for a number of things. If one apple symbol stands for 5 apples, then 4 apple symbols stand for:

- **(A)** 4 apples
- **(B)** 20 apples
- **(C)** 5 apples
- **(D)** 9 apples

29. When you heat the liquid inside a thermometer, the liquid column rises up the tube. This happens because, on heating, the liquid:

- **(A)** disappears
- **(B)** freezes
- **(C)** contracts
- **(D)** expands

30. Tiny insects like a grasshopper do not breathe through a nose. Air enters and leaves their bodies through small openings along the sides called:

- **(A)** nostrils
- **(B)** gills
- **(C)** spiracles
- **(D)** stomata

31. An algebraic expression that has exactly TWO unlike terms - for example 3x + 5 - is called a:

- **(A)** trinomial
- **(B)** variable
- **(C)** monomial
- **(D)** binomial

32. The pulse rates of three friends are 70, 76 and 70 beats per minute. The MODE of this small data set is:

- **(A)** 72
- **(B)** 70
- **(C)** 76
- **(D)** 6

33. On its own, heat moves only from a warmer object towards a cooler one, and this transfer keeps going until the two objects settle at the:

- **(A)** highest temperature
- **(B)** boiling point
- **(C)** freezing point
- **(D)** same temperature

34. Yeast can release energy from sugar even without oxygen. In this oxygen-free (anaerobic) respiration, yeast produces carbon dioxide and:

- **(A)** oxygen
- **(B)** water
- **(C)** lactic acid
- **(D)** alcohol

35. An expression that contains only ONE term, such as 6xy, is called a:

(A) constant
(B) binomial
(C) trinomial
(D) monomial

36. In a bar graph showing rainfall for several months, the month with the MOST rainfall is shown by the bar that is the:

(A) tallest
(B) widest
(C) shortest
(D) first on the left

37. Land and water are both warmed by the Sun, but during the day the LAND heats up faster than the sea. This is because, compared with water, land:

(A) warms up more slowly
(B) cannot be heated
(C) warms up more quickly
(D) stays at one temperature

38. Inside every living cell, food (glucose) is broken down with the help of oxygen to set free the energy stored in it. This energy-releasing process is called:

(A) transpiration
(B) respiration
(C) digestion
(D) germination

39. Look at the expression 5a + 3b + 2a. After combining the like terms, it simplifies to:

(A) 10ab
(B) 7a + 5b
(C) 7a + 3b
(D) 8a + 2b

40. The temperatures recorded on five days were 31, 29, 33, 30 and 32 degrees. Arranged in order, the MEDIAN temperature is:

(A) 31
(B) 33
(C) 32
(D) 29

41. A cook fixes a wooden or plastic handle to a steel frying pan. The reason is that wood and plastic are good insulators, meaning they:

(A) do not let heat pass through easily
(B) make extra heat
(C) turn heat into light
(D) let heat pass through quickly

42. During hard exercise your breathing rate goes UP - you breathe faster and deeper. The main reason for this is that the working muscles need more:

- **(A)** food
- **(B)** carbon dioxide
- **(C)** oxygen
- **(D)** water

43. The perimeter of a square is the total length around it. For a square whose side is s, the perimeter is best written as:

- **(A)** 2s
- **(B)** s^2
- **(C)** 4s
- **(D)** s + 4

44. When you toss a fair coin once, it is equally likely to land heads or tails. The probability (chance) of getting HEADS is:

- **(A)** 1/6
- **(B)** 2
- **(C)** 1/2
- **(D)** 1

45. A clinical thermometer has a sharp BEND, or kink, in its narrow tube just above the bulb. The purpose of this kink is to:

- **(A)** stop the mercury from slipping back on its own
- **(B)** show the boiling point
- **(C)** hold extra mercury
- **(D)** make it heat up faster

46. Plant roots are buried in the soil, yet root cells still need oxygen to respire. They get this oxygen from the:

- **(A)** air present in spaces between soil particles
- **(B)** sunlight falling on the leaves
- **(C)** water flowing past them
- **(D)** carbon dioxide in the soil

47. Two terms are 'unlike terms' when their variable parts are different. Because of this, the terms 4x and 4y:

- **(A)** cannot be added into a single term
- **(B)** add up to 8x
- **(C)** add up to 8xy
- **(D)** add up to 16

48. Out of all the kinds of graph, a double bar graph is the best choice when your aim is to:

- **(A)** show how one value changes over years
- **(B)** show a single set of data
- **(C)** compare two sets of data side by side
- **(D)** find the area of a shape

49. On the Celsius scale used by most thermometers, pure water freezes at 0°C. On the same scale, pure water boils at:

- **(A)** 37°C
- **(B)** 100°C
- **(C)** 50°C
- **(D)** 212°C

50. Putting it all together, the word equation for respiration that uses oxygen (aerobic respiration) is: glucose + oxygen -> carbon dioxide + water + ____. The blank stands for:

- **(A)** sunlight
- **(B)** soil
- **(C)** oxygen
- **(D)** energy

51. To form an expression for 'the cost of buying n pens, when each pen costs 5 rupees', you would write:

- **(A)** 5 + n
- **(B)** 5n
- **(C)** n / 5
- **(D)** n - 5

52. A bag holds an equal number of red and blue balls only. The probability of drawing out a GREEN ball from this bag is:

- **(A)** 1/2
- **(B)** 1
- **(C)** 1/3
- **(D)** 0

53. People living near the sea often find their summers and winters less extreme than people far inland. This milder coastal climate is mainly because water:

- **(A)** heats up and cools down slowly
- **(B)** heats up and cools down quickly
- **(C)** blocks the Sun's heat
- **(D)** never changes temperature

54. Both breathing in and breathing out involve movements of the chest. During breathing OUT (exhalation), the diaphragm:

- **(A)** contracts and moves down
- **(B)** disappears completely
- **(C)** turns into a lung
- **(D)** relaxes and moves up

55. Starting at a room temperature of 20°C, a heater raises the temperature by 2 degrees every hour. The temperature after t hours is given by the expression:

- **(A)** 2 + 20t
- **(B)** 20 + 2t
- **(C)** 20 x 2t
- **(D)** 20 - 2t

56. An impossible event has a probability of 0 and a certain (sure) event has a probability of 1. So the probability of any event must always lie:

- **(A)** above 1 always
- **(B)** between 0 and 1
- **(C)** below 0 always
- **(D)** between 1 and 10

57. A doctor must NOT use an ordinary clinical thermometer to measure the temperature of boiling water. The main reason is that boiling water is:

- **(A)** too cold to measure
- **(B)** far hotter than the clinical thermometer's range
- **(C)** not a liquid
- **(D)** too pure to read

58. Stems of woody plants are covered by bark, which air cannot easily cross. For gas exchange, such stems have small openings in the bark called:

- **(A)** gills
- **(B)** spiracles
- **(C)** stomata
- **(D)** lenticels

59. The coefficient of the variable in the term -5y is the number multiplying y, including its sign. That coefficient is:

- **(A)** y
- **(B)** 5
- **(C)** -5
- **(D)** -1

60. Before you can find the median of a list of marks, the first thing you must always do is:

- **(A)** pick the largest mark
- **(B)** arrange the marks in order of size
- **(C)** count how many marks there are
- **(D)** add up all the marks

61. Heat is best described not as a substance but as a form of:

- **(A)** energy
- **(B)** light
- **(C)** matter
- **(D)** sound

62. A person normally takes about 15 to 18 breaths each minute while at rest. One full 'breath' here means:

- **(A)** ten heartbeats
- **(B)** only one breathing in
- **(C)** one breathing in followed by one breathing out
- **(D)** one yawn

63. Removing the like terms, simplify 9m - 4m. The result is:

- **(A)** 5
- **(B)** $5m^2$
- **(C)** 5m
- **(D)** 13m

64. Five readings are 12, 15, 11, 18 and 14. To work out their RANGE you find:

- **(A)** 18 + 11 = 29
- **(B)** 15 - 14 = 1
- **(C)** 12 - 11 = 1
- **(D)** 18 - 11 = 7

65. Imagine two metal balls, one at 80°C and one at 30°C, touching each other. Heat will flow from the:

- **(A)** ground into both balls
- **(B)** air into the hotter ball
- **(C)** 80°C ball to the 30°C ball
- **(D)** 30°C ball to the 80°C ball

66. Cockroaches, ants and other insects have a network of fine air tubes that carry air from the spiracles to every part of the body. These tubes are called:

- **(A)** nerves
- **(B)** tracheae
- **(C)** lungs
- **(D)** blood vessels

67. The value of the expression $x^2 + 1$ when $x = 3$ is:

- **(A)** 10
- **(B)** 7
- **(C)** 9
- **(D)** 16

68. A 'frequency' in a data table tells you:

- **(A)** the average of the data
- **(B)** the largest value present
- **(C)** how many times a value occurs
- **(D)** how large the value is

69. The metal pot on a stove is made of metal on purpose, while the handle is made of wood. The pot is metal because metals are:

- **(A)** able to make heat
- **(B)** lighter than wood
- **(C)** poor conductors of heat
- **(D)** good conductors of heat

70. Even at night, when there is no sunlight and no photosynthesis, a green plant continues to:

- **(A)** stop all life processes
- **(B)** respire, taking in oxygen and giving out carbon dioxide
- **(C)** take in carbon dioxide only
- **(D)** make food using sunlight

71. Written as an algebraic expression, 'the sum of x and twice y' is:

- **(A)** 2x + y
- **(B)** x + 2y
- **(C)** 2xy
- **(D)** x + y + 2

72. The number of glasses of water a boy drank on six days were 5, 6, 5, 7, 6 and 5. The MODE of this data is:

- **(A)** 5
- **(B)** 7
- **(C)** 3
- **(D)** 6

73. During the daytime at a beach, the cool wind that blows from the sea towards the land is called a:

- **(A)** cyclone
- **(B)** monsoon wind
- **(C)** land breeze
- **(D)** sea breeze

74. In an experiment, soaked seeds are kept in a sealed flask. After a day the air in the flask has less oxygen and more carbon dioxide, and a thermometer shows the flask has grown slightly:

- **(A)** colder
- **(B)** unchanged in temperature
- **(C)** warmer
- **(D)** frozen

75. In a classroom each bench seats 2 students. An algebraic expression for the number of students seated on b benches is:

- **(A)** 2b
- **(B)** b / 2
- **(C)** 2 - b
- **(D)** b + 2

76. On a bar graph the vertical scale is marked 0, 10, 20, 30 and so on. A bar that reaches exactly halfway between the 20 and 30 lines stands for a value of:

- **(A)** 50
- **(B)** 25
- **(C)** 23
- **(D)** 30

77. Why does a metal chair feel COLDER to touch than a wooden chair in the same cool room, even though both are at the same temperature?

- **(A)** wood makes its own heat
- **(B)** metal conducts heat away from your hand faster
- **(C)** the metal is actually at a lower temperature
- **(D)** metal reflects cold onto your hand

78. A tadpole living in water breathes with gills, but when it grows into an adult frog living partly on land, it breathes mainly with its:

- **(A)** lungs (and moist skin)
- **(B)** gills only, like the tadpole
- **(C)** spiracles like an insect
- **(D)** roots

79. An algebraic expression made up of THREE unlike terms, such as $x + y + 7$, is called a:

- **(A)** binomial
- **(B)** trinomial
- **(C)** monomial
- **(D)** coefficient

80. A weather station noted the day's temperature as 27°C, 27°C, 29°C, 31°C and 26°C over five days. The mean temperature is:

- **(A)** 31°C
- **(B)** 28°C
- **(C)** 27°C
- **(D)** 140°C

81. A thermometer works by linking temperature to the length of a thin column of liquid. The liquid traditionally used in a clinical thermometer is:

- **(A)** water
- **(B)** mercury
- **(C)** oil
- **(D)** milk

82. Compared with the fresh air we breathe in, the air we breathe OUT contains:

- **(A)** only carbon dioxide
- **(B)** more oxygen and less carbon dioxide
- **(C)** less oxygen and more carbon dioxide
- **(D)** no oxygen at all

83. A 'term' in an algebraic expression is a single part separated from the others by a plus or minus sign. The expression 4x - 3y + 7 has how many terms?

- **(A)** two
- **(B)** four
- **(C)** one
- **(D)** three

84. The data 8, 3, 5, 9 and 5 is to be summarised by its MEDIAN. After ordering the values, the median is:

- **(A)** 9
- **(B)** 3
- **(C)** 5
- **(D)** 8

85. When a substance is heated, its tiny particles gain energy and move about more. Because of this extra movement, most substances on heating:

- **(A)** contract (take up less space)
- **(B)** turn into a gas at once
- **(C)** become heavier
- **(D)** expand (take up more space)

86. The opening and closing of a stoma (a leaf pore) is controlled by a pair of specially shaped cells on either side of it. These are the:

- **(A)** blood cells
- **(B)** guard cells
- **(C)** root hairs
- **(D)** nerve cells

87. Simplify the expression by collecting like terms: 3p + 2q + 5p + q. The simplest form is:

- **(A)** 8p + 3q
- **(B)** 11pq
- **(C)** 8p + 2q
- **(D)** 10p + 3q

88. A coin is tossed and a single dice is rolled by two different children. The child rolling the DICE has a smaller chance of getting any one chosen number because:

- **(A)** the coin can land on its edge
- **(B)** the dice is heavier than the coin
- **(C)** the dice has more equally likely outcomes than the coin
- **(D)** a dice cannot be fair

89. The actual amount of heat ENERGY a body contains depends not only on its temperature but also on how much of the substance there is. So a large bucket of warm water contains more heat than a small cup of water at the same temperature because the bucket has:

- **(A)** no heat at all
- **(B)** colder water
- **(C)** more water
- **(D)** a higher temperature

90. Some bacteria can live and respire in places with no oxygen at all, such as deep in waterlogged mud. Such organisms are described as:

- **(A)** herbivorous
- **(B)** nocturnal
- **(C)** anaerobic
- **(D)** aerobic

91. Reading the statement carefully, the expression for 'seven less than three times a number n' is:

- **(A)** $3n + 7$
- **(B)** $3(n - 7)$
- **(C)** $3n - 7$
- **(D)** $7 - 3n$

92. A bar graph shows the number of books read by four children: Asha 6, Ravi 4, Meena 8 and Sam 6. The total number of books read by all four is:

- **(A)** 6
- **(B)** 24
- **(C)** 18
- **(D)** 8

93. After a hot meal is served, a steel bowl of soup cools down while the cooler air around it warms up slightly. Eventually the soup stops cooling when it reaches:

- **(A)** 0°C, the freezing point
- **(B)** 100°C, the boiling point
- **(C)** its own original temperature
- **(D)** the temperature of the surrounding air (room temperature)

94. A simple way to remember the role of breathing is that it supplies the body with oxygen and removes a waste gas. That waste gas removed by breathing is:

- **(A)** water
- **(B)** nitrogen
- **(C)** oxygen
- **(D)** carbon dioxide

95. If a number is called x, then 'the number that is 1 more than double x' is written as:

- **(A)** $2x + 1$
- **(B)** $2(x + 1)$
- **(C)** $x + 2$
- **(D)** $2x - 1$

96. Six students were asked how many glasses of water they drank: the values were 4, 6, 5, 7, 5 and 3. Because there are an EVEN number of values, the median is found by:

- **(A)** adding all six values
- **(B)** averaging the two middle values once the data is ordered
- **(C)** choosing the largest value
- **(D)** taking the single middle value

97. In summary, the single best one-word description of what a thermometer actually measures is:

- **(A)** temperature
- **(B)** weight
- **(C)** heat energy
- **(D)** pressure

98. Tying the chapter together: the chief PURPOSE of respiration in any living organism is to:

- **(A)** release energy from food for life processes
- **(B)** remove extra water from the body
- **(C)** make food using sunlight
- **(D)** help the organism grow taller only

99. Putting respiration and algebra together: if a resting person takes 16 breaths a minute, the number of breaths in 5 minutes is found by evaluating 16m at m = 5, which gives:

- **(A)** 80
- **(B)** 21
- **(C)** 11
- **(D)** 16

100. Finally, combining data handling with respiration: the breathing rates of five classmates are 18, 20, 16, 22 and 19 breaths per minute. Their MEAN breathing rate is:

- **(A)** 19
- **(B)** 95
- **(C)** 16
- **(D)** 22